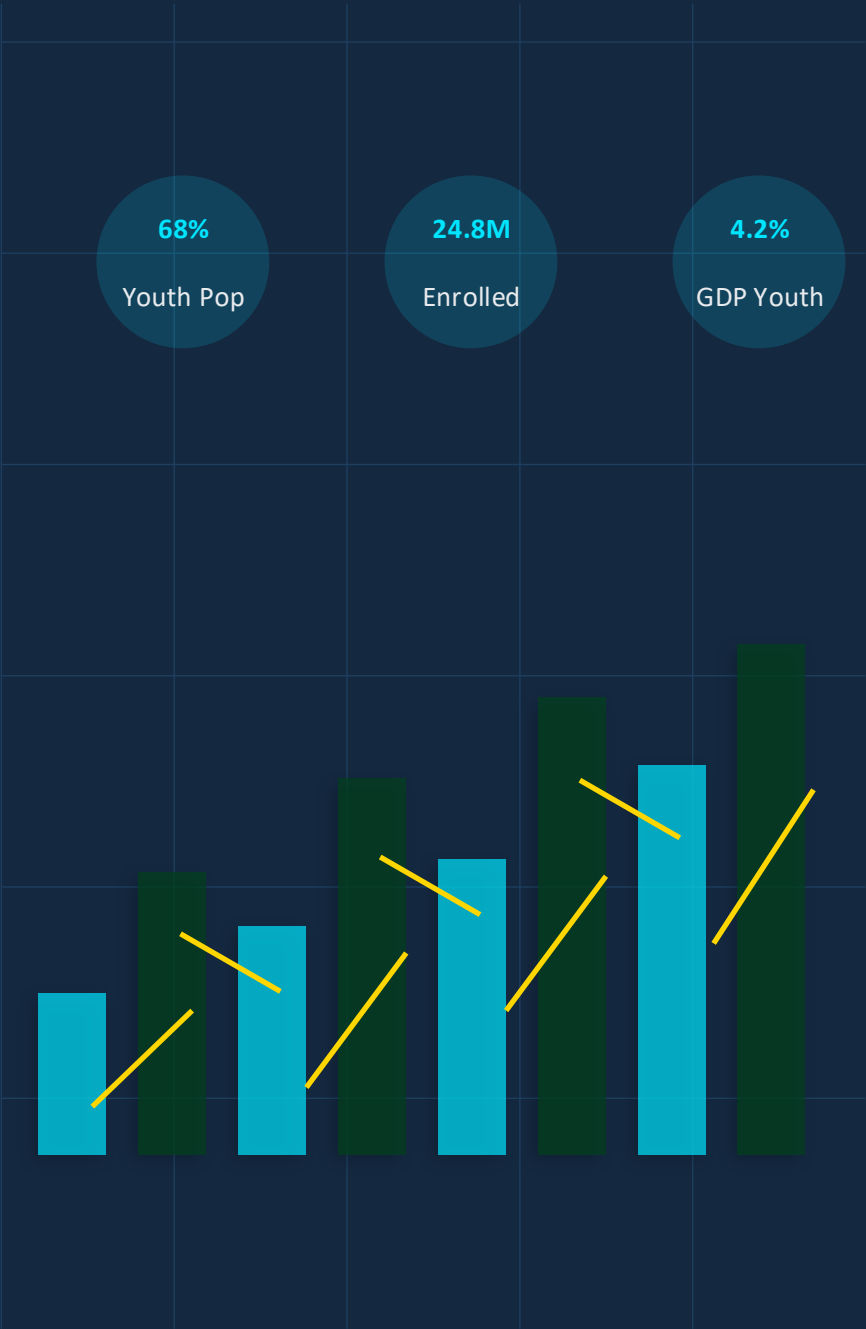


Empowering Pakistan Through Youth Analytics & Data-Driven Nation Building

Muhammad Ehtisham 22i-0695
Rayyan Asim 23i-5002
Hussain Asad 23i-5085

A DATA ANALYTICS & VISUALIZATION PERSPECTIVE ON PAKISTAN

INSTRUCTOR		UNIVERSITY	DATE
Dr. Nadeem Ahmed		FAST NU, ISB	15 May 2026
SDG 4 Education	SDG 8 Dec. Work	SDG 9 Innovation	SDG 16 Institutions



Introduction: Nation Building Through Analytics



Nation Building

Systematic process of constructing national identity, institutions, and economic capacity for sustainable development.



Youth as Economic Engine

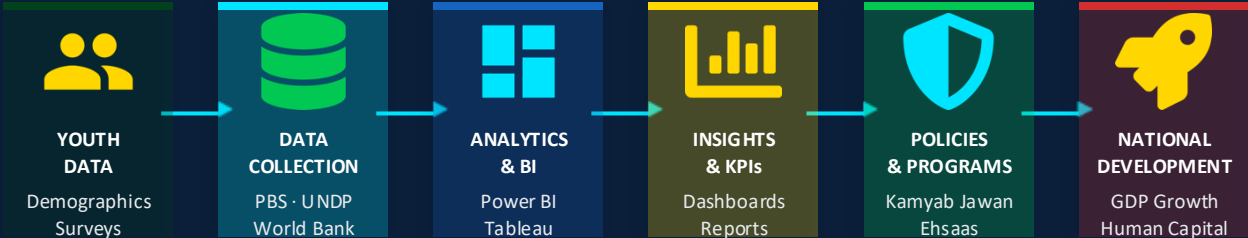
Youth constitute 64% of Pakistan's population — the largest demographic dividend in the nation's history.



Analytics in Policymaking

Data-driven governance replaces intuition with evidence, enabling targeted, cost-effective interventions.

ANALYTICS PIPELINE FOR NATIONAL DEVELOPMENT

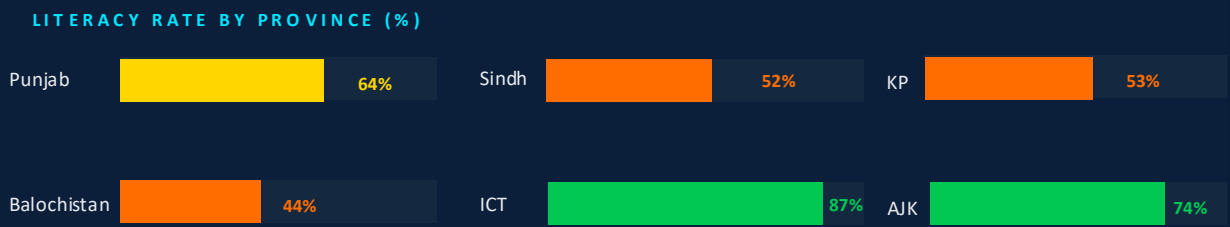
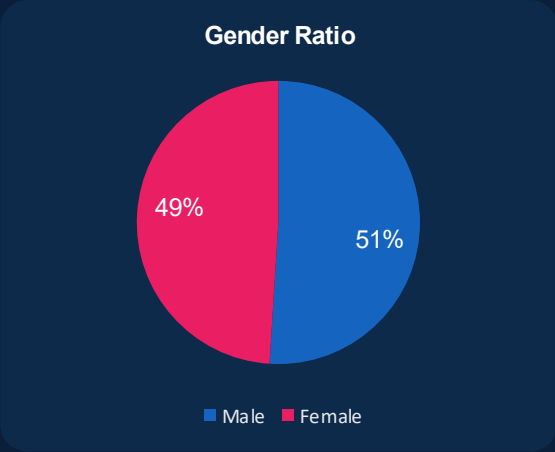
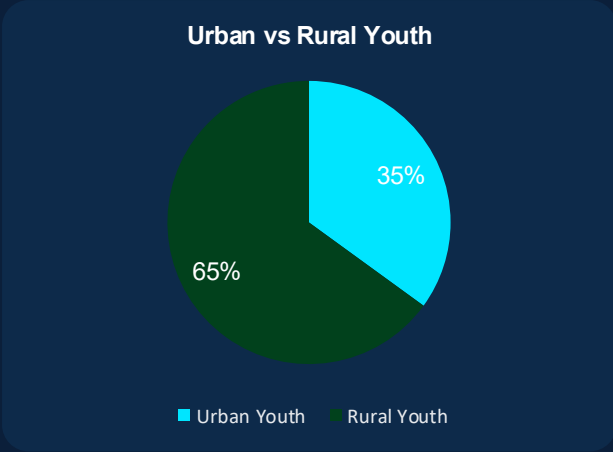
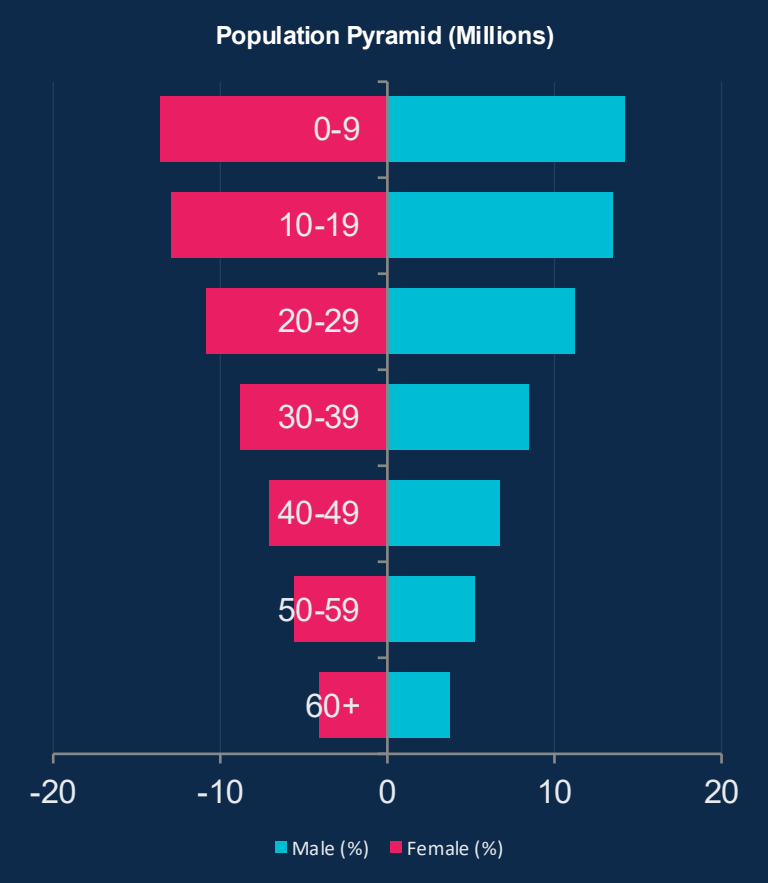


 Evidence-Based Policy 3x better outcomes	 Resource Efficiency 40% cost reduction	 Targeted Interventions Provincial precision
 Real-Time Monitoring Live dashboards	 Predictive Planning 10-year forecasts	 Stakeholder Trust Transparent data

Sources: World Bank (2024); UNDP Human Development Report (2024); PBS Pakistan (2023)

Pakistan Youth Demographics Dashboard

POWER BI-STYLE ANALYTICS DASHBOARD | DATA: PBS 2023-2024 | UPDATED: Q1 2025



KEY INSIGHT

"Pakistan's demographic dividend can become an economic advantage through data-driven governance."

Pakistan ranks 3rd globally by youth population size

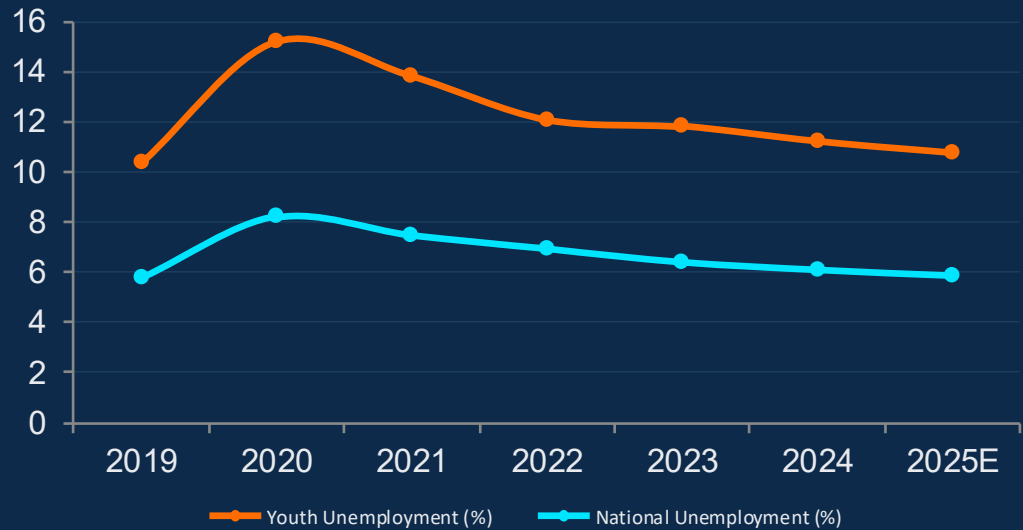
Source: PBS Census 2023
UNDP Pakistan 2024
World Bank Data

Sources: Pakistan Bureau of Statistics (2023); UNDP Pakistan Human Development Report (2024); World Bank Open Data (2024); UNICEF Pakistan (2024)

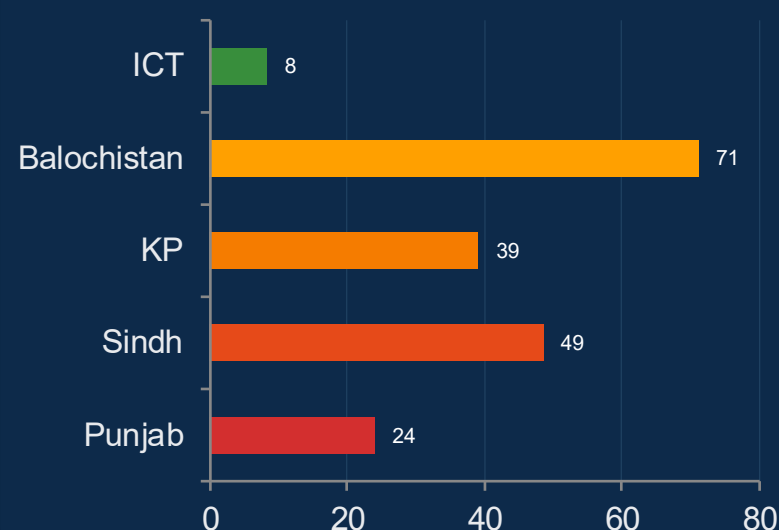
Data Analytics in Understanding Youth Issues

BUSINESS INTELLIGENCE · RISK SEGMENTATION · PREDICTIVE TREND ANALYSIS

Youth vs National Unemployment Trend



Poverty Rate by Province (%)



NEET Youth

28%

Not in Education, Employment or Training

▲ 3.2% YoY

Digital Divide

53%

Rural youth with no internet access

▼ 4.1% YoY

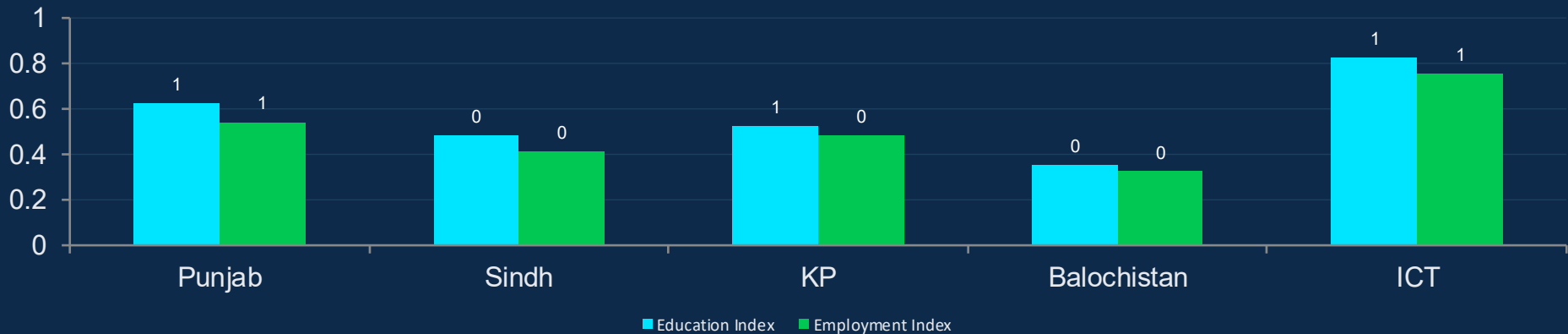
Out-of-School

22.8M

Children aged 5-16 not enrolled

▼ 1.5M since 2022

Regional Development Indices by Province (0-1 Scale) — Source: UNDP HDI 2024



Sources: PBS Labour Force Survey (2023-24); UNDP Human Development Report Pakistan (2024); World Bank Poverty Data (2024)

Youth Employment Analytics

PREDICTIVE FORECASTING · FREELANCING GROWTH · IT EXPORTS · STARTUP ECOSYSTEM

11.8%

Youth Unemployment

\$3.5B

IT Exports 2024

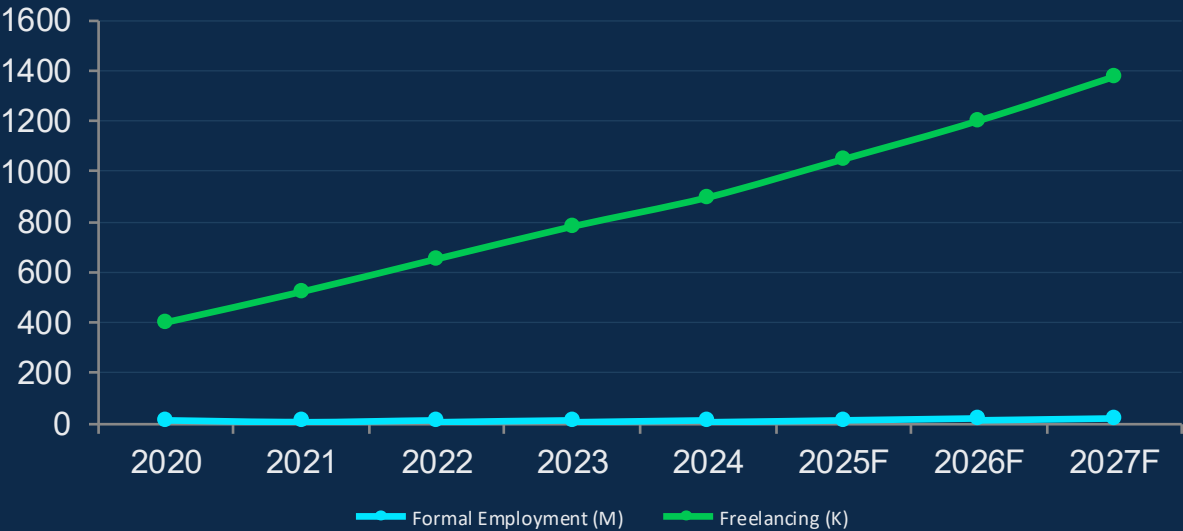
900K+

Freelancers

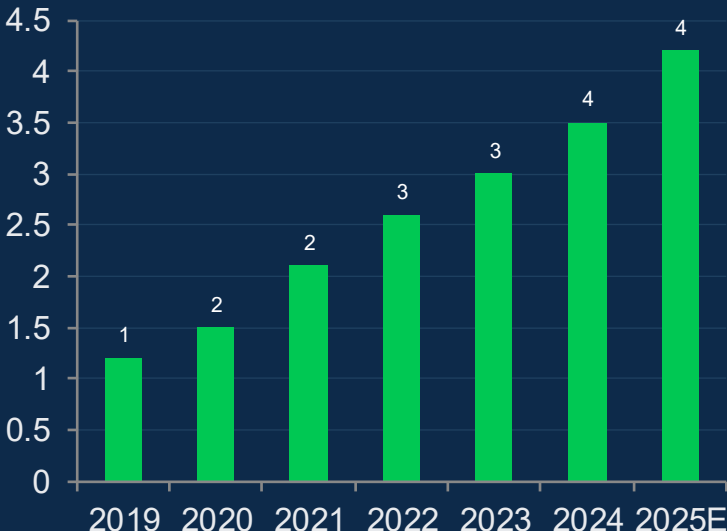
700+

Active Startups

Youth Employment Forecast (Formal + Freelancing) 2020-2027F



Pakistan IT Exports Growth (USD Billion)



ANALYTICS TOOLS IN USE:

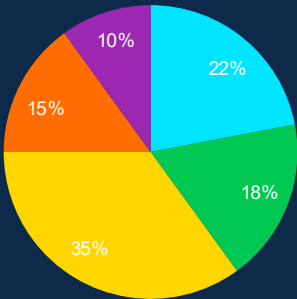
Power BI

Tableau

Excel Analytics

Python/R

Employment Sectors



Sources: State Bank of Pakistan (2024); PSEB IT Exports Report (2024); PBS Labour Force Survey (2024); Payoneer Freelancer Report (2024)

Education Analytics & Skill Development

SDG 4 – QUALITY EDUCATION | LEARNING ANALYTICS | SKILL GAP ANALYSIS

58%

National Literacy

32%

Female Rural Literacy

22.8M

Out-of-School Children

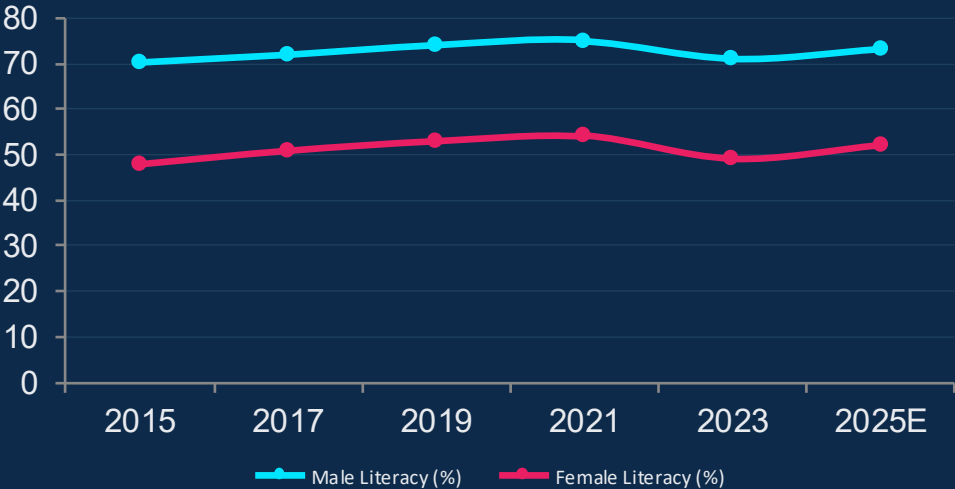
2.9%

GDP on Education

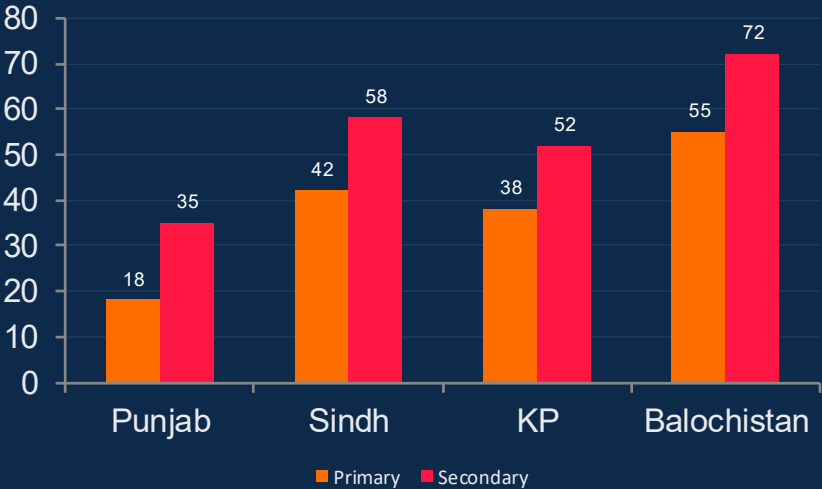
68%

Primary Completion

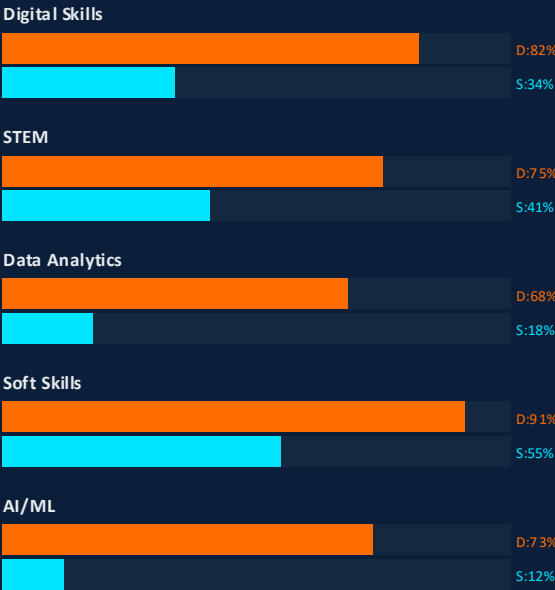
Gender Literacy Gap Trend 2015-2025



School Dropout Rates by Province & Level (%)



SKILL GAP ANALYSIS (Demand vs Supply %)



🇵🇰 HEC ANALYTICS: 182 Universities | 1.9M Enrolled Students | 38% Female Enrollment | 65% STEM Gap | DigiSkills: 4.2M Beneficiaries (2024)

Sources: HEC Pakistan Annual Report (2024); UNICEF Education Pakistan (2024); PBS PSLM Survey (2023); World Bank Education Data (2024)

Social Media & Digital Analytics

BIG DATA VISUALIZATION | ENGAGEMENT ANALYTICS | MOBILE PENETRATION DASHBOARD

82M+

Mobile Users

47%

Internet Penetration

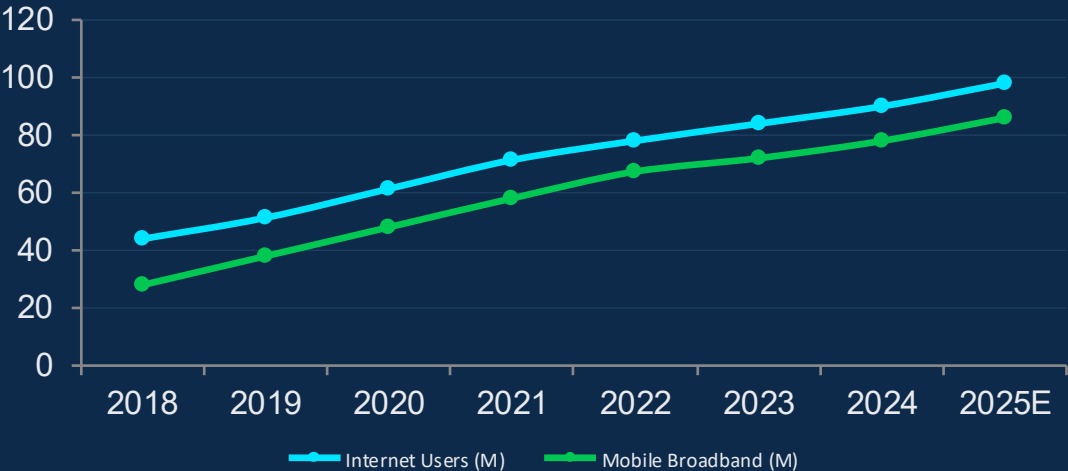
43M

Social Media Users

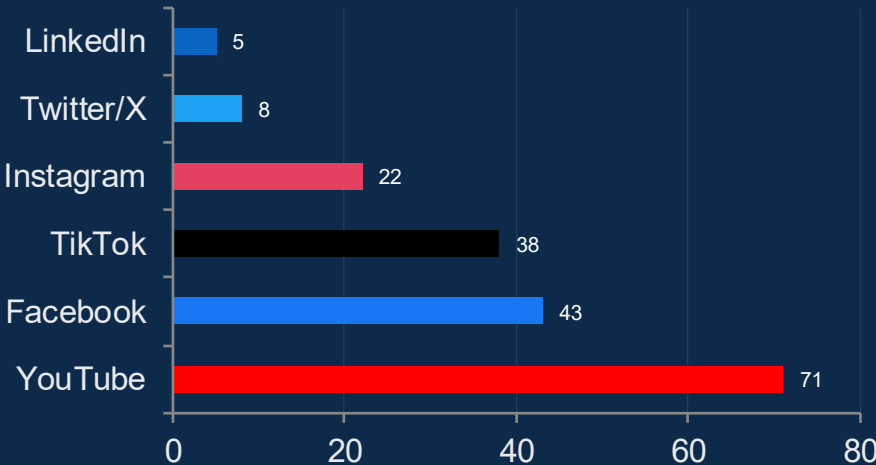
71%

Youth on Smartphones

Internet & Mobile Broadband Growth



Social Media Platform Usage — Pakistan Youth 2024



DigiSkills Enrollments

4.2M

Cumulative 2024

Coursera Users

850K

Pakistani learners

YouTube Learning

68%

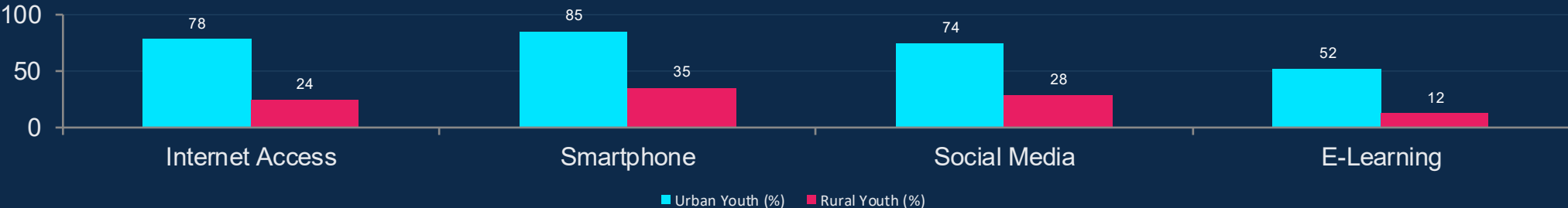
Youth use for study

E-Learning Growth

340%

Post-COVID surge

Urban vs Rural Digital Access Gap — Analytics KPI Dashboard | Source: PTA, GSMA 2024



Sources: PTA Annual Report (2024); GSMA Mobile Economy (2024); DataReportal Digital Pakistan (2024); HEC DigiSkills (2024)

Youth Entrepreneurship & Innovation Analytics

STARTUP ECOSYSTEM | INNOVATION KPIs | INVESTMENT TRENDS | SDG 9 INNOVATION

700+

Active Startups

\$450M

VC Investment 2024

23

NICs Nationwide

4.2M

DigiSkills Grads

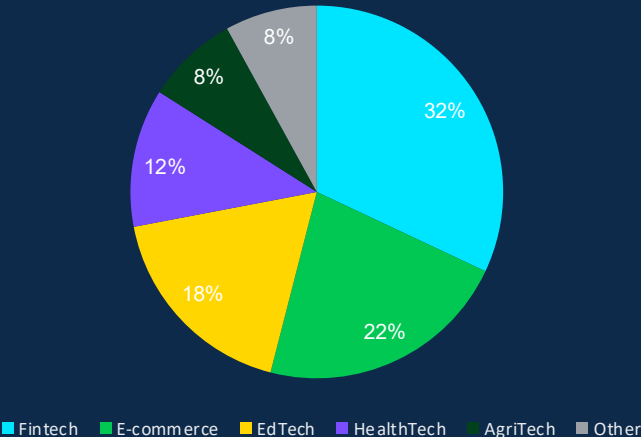
\$3.5B

IT Export Revenue

Pakistan Startup VC Investment Growth (USD Million)



Startup Ecosystem by Sector (2024)



IGNITE PAKISTAN

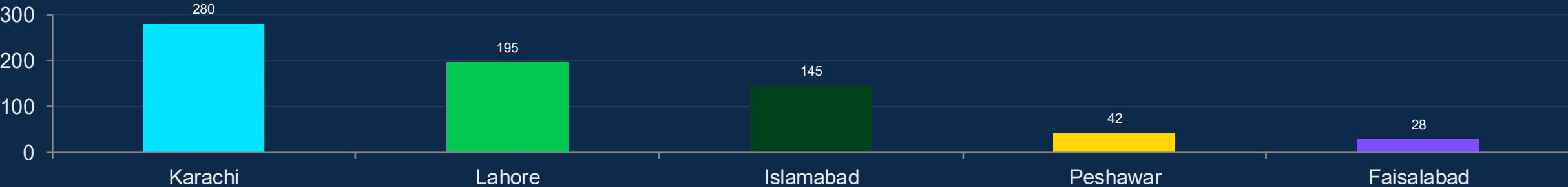
National ICT R&D Fund
700+ tech projects
funded since 2009

DIGISKILLS.PK

4.2M+ graduates
29 technical courses
Free for all Pakistanis

NIC NETWORK

Innovation Hub Distribution — Active Startups by City (2024) | Source: Invest2Innovate Pakistan



Sources: Invest2Innovate Pakistan Ecosystem Report (2024); PSEB (2024); DigiSkills.pk (2024); IGNITE Pakistan (2024); SBP Fintech Report (2024)

Challenges Faced by Pakistani Youth

⚠️ RISK ANALYTICS DASHBOARD | CRITICAL INDICATORS | PREDICTIVE RISK ASSESSMENT



6M+

Brain Drain (2015-2024)



38.3%

Below Poverty Line



28%

NEET Rate



72%

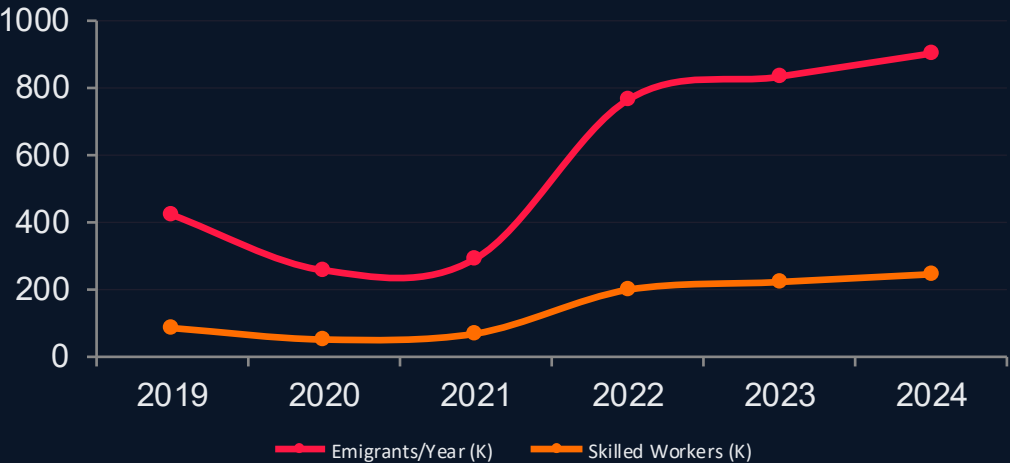
Gender Gap Score



47%

Mental Health Issues

Brain Drain: Emigration Trend 2019-2024



ROOT CAUSE ANALYSIS — SEVERITY INDEX (0-100)

Unemployment

Skill mismatch, economic contraction

Brain Drain

Better opportunities abroad

Education Gap

Underfunding, rural access

Gender Inequality

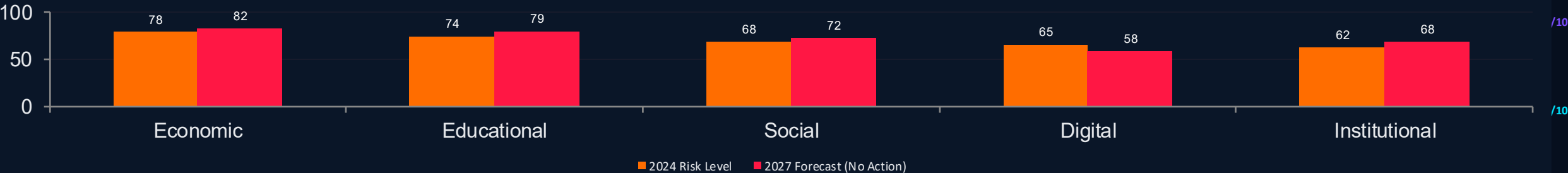
85/100

78/100

82/100

74/100

Risk Level Forecast 2024 vs 2027 — Predictive Model (Without Policy Intervention)



Sources: Bureau of Emigration & Overseas Employment (2024); PBS Poverty Survey (2024); WHO Mental Health Atlas (2024); WEF Global Gender Gap Report (2024)

Gender Analytics & Women Empowerment

SDG 5 – GENDER EQUALITY | WORKFORCE ANALYTICS | PROVINCIAL DISPARITY DASHBOARD

49%

Female Literacy

22%

Female Workforce

143rd

Gender Gap Index Rank

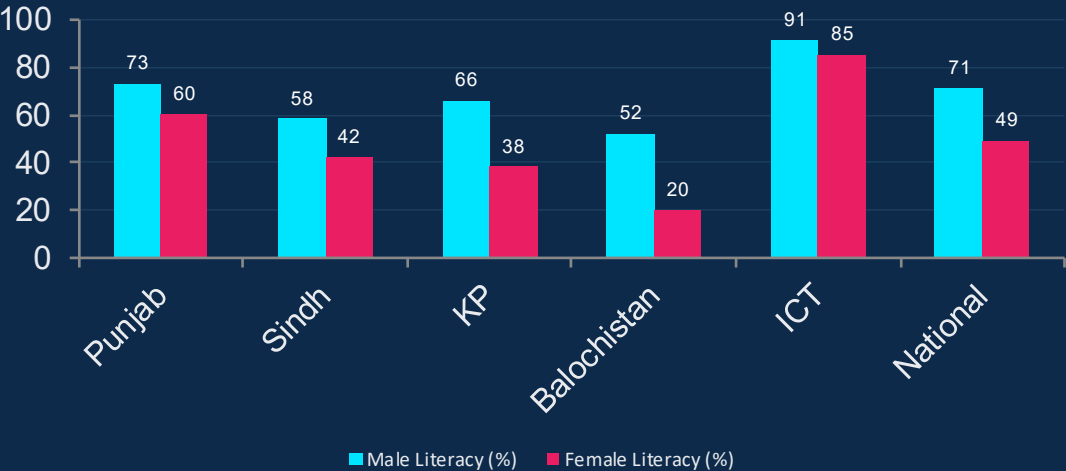
38%

Female University Enrollment

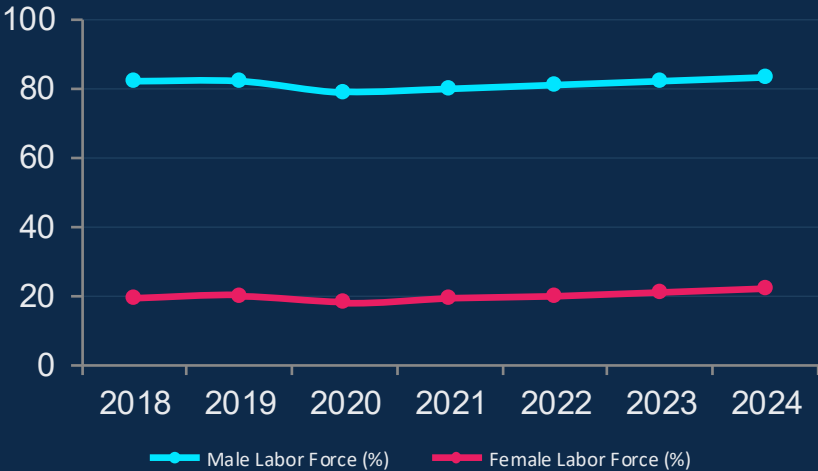
58%

Wage Gap

Male vs Female Literacy Rate by Province (%)



Labor Force Participation Trend by Gender (%)



SDG 5 Progress

37%

Gender parity achieved

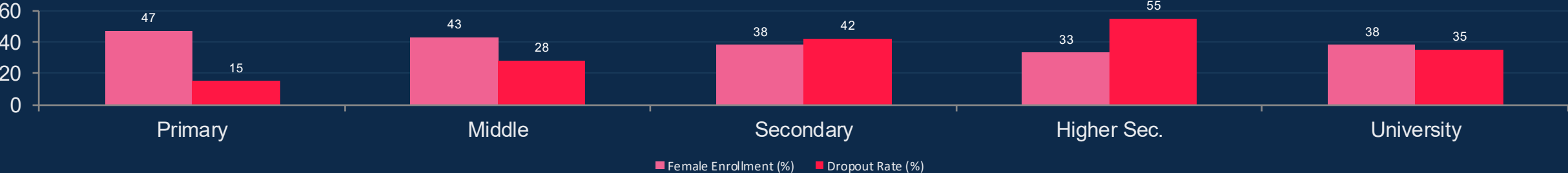
Girls Education

+12%

Enrollment since 2019

Women Entrepreneurs

Female Education Funnel — Enrollment vs Dropout Rates (%) by Level | Analytics Alert: 55% dropout at Higher Secondary



Sources: WEF Global Gender Gap Report (2024); PBS PSLM (2023); ILO Labour Force Statistics (2024); UNICEF MICS Pakistan (2024)

Government Youth Programs Analytics Dashboard

PROGRAM PERFORMANCE | BENEFICIARY ANALYTICS | ROI ASSESSMENT | SDG 16

PM Youth Program

REACH
4.2M

BUDGET
PKR 40B

ROI: **3.2x**



Kamyab Jawan

REACH
630K

BUDGET
PKR 100B loans

ROI: **2.8x**



Ehsaas Program

REACH
15M+

BUDGET
PKR 480B

ROI: **4.1x**



DigiSkills.pk

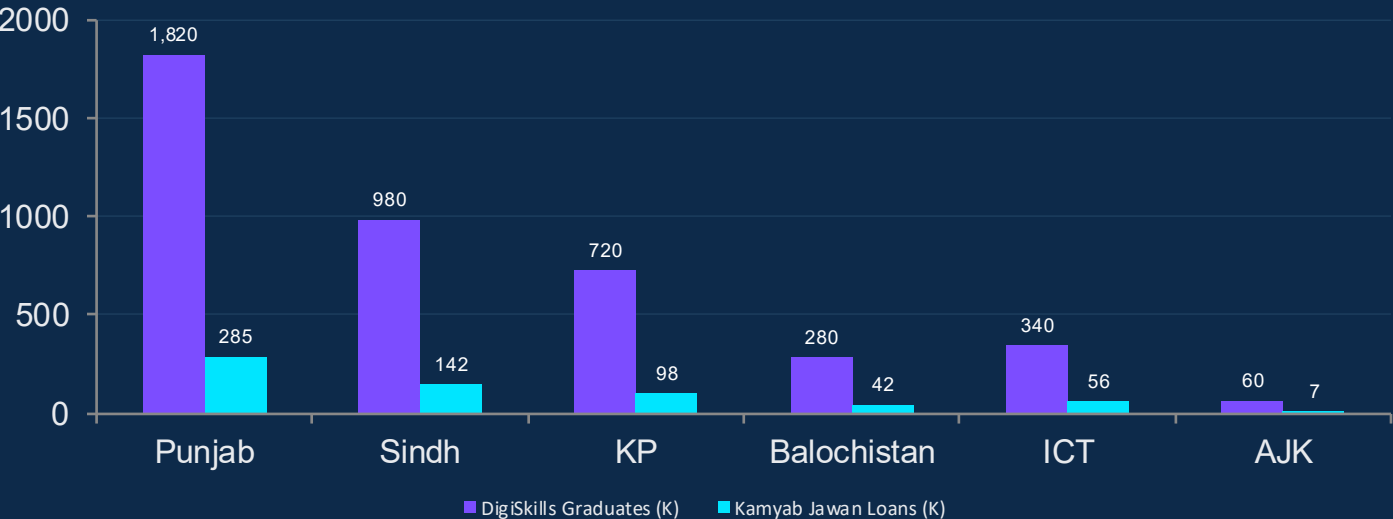
REACH
4.2M

BUDGET
PKR 4B

ROI: **8.5x**



Geographic Distribution of Program Beneficiaries by Province



PROGRAM ROI SCORECARD



Sources: Government of Pakistan Program Reports (2024); BISPA Annual Review (2024); DigiSkills.pk Impact Report (2024); Planning Commission Pakistan (2024)

Predictive Analytics for Pakistan's Future

AI-POWERED FORECASTING | FUTURE SKILLS DASHBOARD | WORKFORCE DEMAND MODELING

85M

New Jobs Needed by 2030

68%

Jobs at Risk from AI

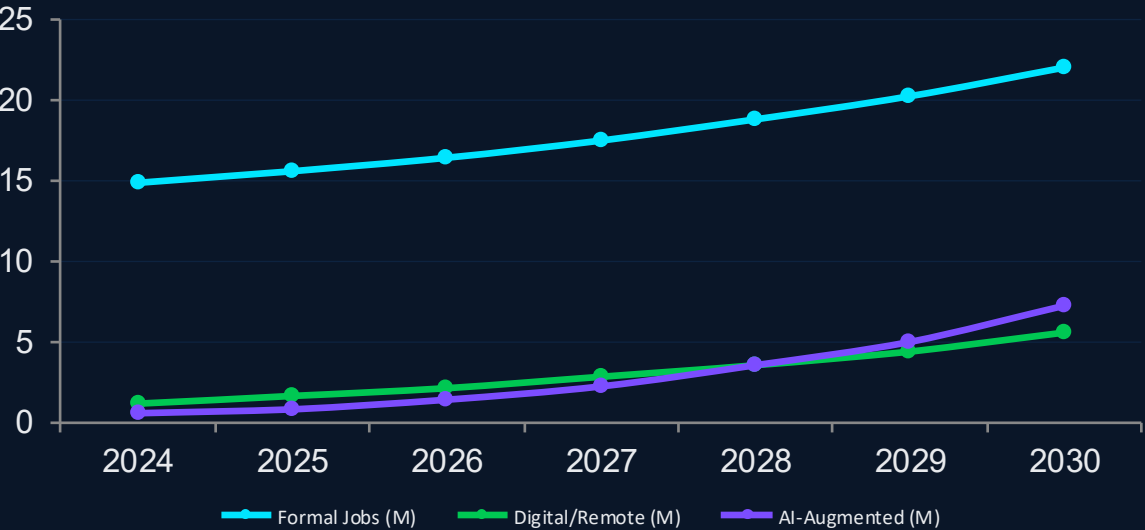
40%

GDP Growth Potential

\$15B

IT Export Target 2027

Youth Employment Forecast Model 2024-2030 (Millions)



FUTURE SKILLS IN DEMAND — 2025-2030



Jobs Automated

2.5M

Routine tasks by 2030

New AI Jobs

3.8M

Net positive for youth

Productivity Gain

+42%

AI-augmented workforce



PREDICTIVE ROADMAP: PAKISTAN DIGITAL ECONOMY



2025

\$4.2B IT Exports



2026

5G Nationwide



2027

1M AI-trained Youth



2028

\$10B Digital Economy



2030

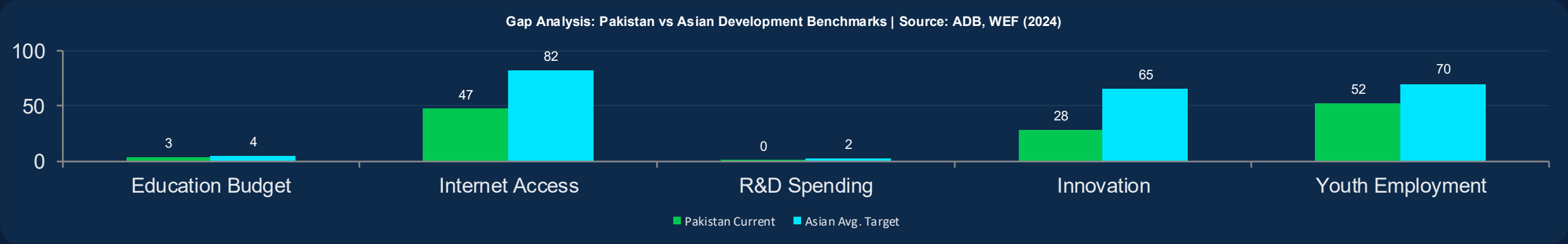
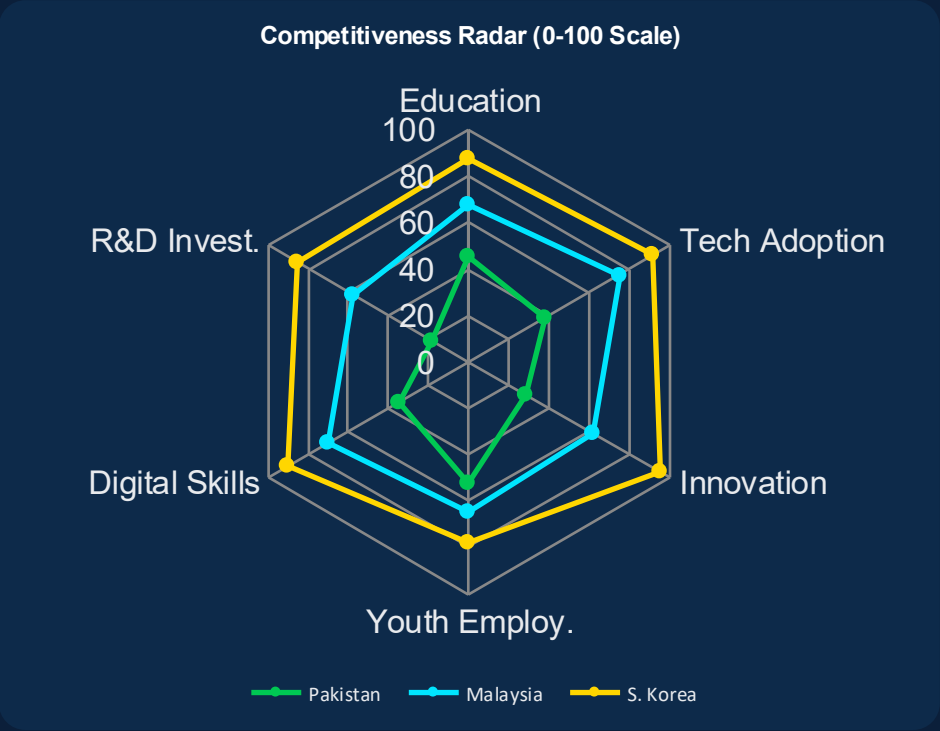
Top 20 Digital Nation

Sources: WEF Future of Jobs Report (2025); McKinsey Global Institute (2024); World Bank Pakistan Digital Economy (2024); Kaspersky AI Research (2024)

Comparative Analytics: Pakistan vs Developed Nations

BENCHMARKING DASHBOARD | RADAR ANALYSIS | COUNTRY COMPARISON MATRIX

Indicator	Pakistan	Malaysia	China	S. Korea	Singapore
Education Investment (% GDP)	2.9%	4.2%	4.6%	5.1%	3.0%
Youth Unemployment (%)	11.8%	10.2%	13.3%	11.2%	5.0%
Internet Penetration (%)	47%	90%	78%	97%	99%
R&D Spending (% GDP)	0.2%	1.4%	2.4%	4.9%	1.9%
Innovation Index Rank	84th	36th	12th	10th	8th
PISA Science Score	422	453	590	541	551
Digital Competitiveness	Low	High	High	V.High	V.High



Sources: WEF Global Competitiveness Report (2024); ADB Key Indicators (2024); OECD PISA (2023); IMD World Digital Competitiveness (2024)

Data Visualization & Evidence-Based Decision-Making

GOVERNMENT DASHBOARDS | REAL-TIME BI SYSTEMS | NATIONAL DATA PORTALS

Power BI

Government KPI Dashboards

43% Market share in govt.

Tableau

Development Agency Reports

31% Used by World Bank

Python/R

Statistical Modeling

18% Academic analytics

Excel BI

Field Data Collection

8% District level use

EVIDENCE-BASED POLICYMAKING ANALYTICS CYCLE



Pakistan Data Portal | data.gov.pk

PBS Open Data | pbs.gov.pk

UNDP Pakistan Analytics | pk.undp.org

SBP Statistics Portal | sbp.org.pk

HEC Analytics Dashboard | hec.gov.pk

BISP MIS Dashboard | bisp.gov.pk

Sources: Government of Pakistan Digital Policy (2024); UNDP Analytics Initiative; Planning Commission Pakistan; WEF Data Governance (2024)

Strategic Recommendations Using Analytics

CONSULTING ROADMAP | SMART GOVERNANCE | DATA-DRIVEN POLICY FRAMEWORK

01

AI-Driven Education Planning

SDG 4

- Deploy predictive analytics for early dropout detection
- Personalized learning dashboards for 22.8M out-of-school children
- Province-specific education investment KPIs

02

Digital Literacy Expansion

SDG 9

- Scale DigiSkills to 10M graduates by 2027
- Rural mobile internet subsidy program
- Digital literacy as national KPI tracked monthly

03

Youth Skills National Database

SDG 8

- Centralized talent analytics platform
- Skills demand-supply matching engine
- Real-time labor market intelligence dashboard

04

Employment Forecasting System

SDG 8

- Provincial employment prediction models
- Sector-specific hiring trend dashboards
- Youth unemployment early warning system

05

Public Sector Analytics Adoption

SDG 16

- Mandatory BI dashboards for all ministries
- National analytics center of excellence
- Open data policy for all government datasets

Sources: Planning Commission of Pakistan (2024); WEF Analytics Governance (2024); UNDP Digital Public Goods (2024); McKinsey Digital Pakistan (2024)

Future Vision: Smart Youth, Smart Pakistan

2030 DIGITAL PAKISTAN VISION | AI GOVERNANCE | INNOVATION ECOSYSTEM



SMART CITIES

5 Smart City pilots
by 2027

IoT sensors
1M+

Real-time city
analytics



AI GOVERNANCE

AI policy
dashboards

Predictive
service delivery

Digital ID
& e-services



DIGITAL ECONOMY

\$15B IT exports
by 2027

5M digital
workers

Fintech
\$5B by 2026



INNOVATION HUBS

50 NICs
nationwide

2,000+
Startups

STEM
500K graduates



DATA NATION

National data
platform

Open data
for 500 datasets

Analytics
for all ministries

PAKISTAN 2030 ANALYTICS TARGETS

80%

Literacy Rate

85%

Internet Access

7%

Youth Unemployment

\$15B

IT Exports

Top 50

Innovation Index

5%

GDP on Education

Conclusion

01 Youth as Strategic Asset

Pakistan's 64% youth population represents the world's most significant untapped demographic dividend — analytics reveals a potential 2-3% additional GDP growth if effectively leveraged through evidence-based policies.

02 Analytics Transforms Governance

Data analytics and business intelligence tools (Power BI, Tableau, Python) convert raw social data into actionable national solutions — enabling precision policy targeting at the district level.

03 Visualization Improves Decisions

Dashboard-driven governance enables real-time KPI monitoring, cross-ministerial coordination, and transparent accountability — improving policy ROI by an estimated 3-4x versus intuition-based decisions.

04 Evidence-Based Sustainable Development

SDG-aligned analytics frameworks ensure Pakistan's youth programs are measurable, comparable, and sustainable — connecting local interventions to global development benchmarks.

SDG 4 • Quality Education

SDG 5 • Gender Equality

SDG 8 • Decent Work

SDG 9 • Innovation

SDG 16 • Strong Institutions

"The youth of today are the leaders of tomorrow — and data is the language through which they will build the Pakistan of their dreams."

— Youth Analytics for National Development, 2025

References

APA FORMAT | ACADEMIC & INSTITUTIONAL SOURCES

INTERNATIONAL ORGANIZATIONS

- Asian Development Bank (ADB). (2024). Key Indicators for Asia and the Pacific 2024. ADB Publications.
- World Bank Group. (2024). Pakistan Development Update: Restoring Stability. World Bank Washington D.C.
- International Labour Organization (ILO). (2024). World Employment and Social Outlook k: Trends 2024. ILO Geneva.
- World Economic Forum (WEF). (2024). Global Competitiveness Report 2024. WEF Geneva.
- United Nations Development Programme (UNDP). (2024). Human Development Report Pakistan 2024. UNDP Islamabad.
- World Economic Forum (WEF). (2025). Future of Jobs Report 2025. WEF Geneva.
- UNICEF Pakistan. (2024). Multiple Indicator Cluster Survey (MICS) Pakistan 2023-24. UNICEF.

PAKISTAN GOVERNMENT & NATIONAL SOURCES

- Bureau of Emigration and Overseas Employment. (2024). Annual Report 2023-24. Government of Pakistan.
- Pakistan Software Export Board (PSEB). (2024). IT & ITeS Industry Annual Report 2024. PSEB.
- Higher Education Commission (HEC) Pakistan. (2024). HEC Annual Report 2023-24. HEC Islamabad.
- Pakistan Telecommunication Authority (PTA). (2024). Annual Report 2023-24. PTA Islamabad.
- Pakistan Bureau of Statistics (PBS). (2023). Population Census 2023: Pakistan at a Glance. PBS.
- Planning Commission of Pakistan. (2024). Pakistan Vision 2025 Progress Report. Government of Pakistan.
- Pakistan Bureau of Statistics (PBS). (2024). Labour Force Survey 2023-24. PBS Islamabad.
- State Bank of Pakistan (SBP). (2024). Annual Report 2023-24: State of Pakistan's Economy. SBP.

RESEARCH & ANALYTICS SOURCES

- DataReportal. (2024). Digital 2024: Pakistan. Kepios Analysis. <https://datareportal.com/reports/digital2024-pakistan>
- OECD. (2023). PISA 2022 Results: Learning Recovery After COVID-19. OECD Paris.
- GSMA Intelligence. (2024). The Mobile Economy Pakistan 2024. GSMA London.
- Payoneer. (2024). Global Freelancer Income Report 2024. Payoneer Inc.
- Invest2Innovate. (2024). Pakistan Startup Ecosystem Report 2024. Islamabad.
- Statista. (2024). Pakistan — Statistics and Data. <https://www.statista.com/topics/2503/pakistan/>
- McKinsey Global Institute. (2024). Digital Pakistan: Transforming the economy through technology. McKinsey & Company.

🚩 *Note: All statistics cited are from official government and international organization sources. Data represents latest available figures (2023-2025). For the most current data, visit respective official portals.*